

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA0307	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	BE-CSE-AI	Date:	

Code of Conduct

*I, **S. Muskan(192472182)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **S. Muskan**

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as “analyzing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

2. An AI-based sentiment analysis platform receives product reviews containing words like “disagree”, “agreement”, and “agreeable”. The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

4. A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.

Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.

Analyze how prefix addition and suffix transformation affect meaning and word class.

Apply the process to decompose each word and map them to a unified base representation.

Derive the morphological structure and normalized root for each input word.

5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.

Identify suffix patterns and map all variations to a common base form without altering meaning.

Apply the process to extract root forms and classify each transformation.

Determine the morphological breakdown and normalized output for each word.

Q1. Morphological Processing of Analyzing, Analysis, Analytical

Scenario Understanding

A search engine receives different word forms such as **analyzing**, **analysis**, and **analytical**. These words have similar meanings but different morphological structures. To improve search accuracy, they should be normalized to a common base representation.

Morphological Processing Approach

1. Tokenize the input word.
2. Identify the root and affixes.
3. Classify each affix as **inflectional** or **derivational**.
4. Remove inflectional endings.
5. Map derivational forms to a common root (lemma).
6. Store the normalized form for indexing.

Morphological Analysis

Word	Morphological Structure	Root	Affix	Type	Normalized Form
analyzing	analyze + -ing	analyze	-ing	Inflectional (Present Participle)	analyze
analysis	analys(e) + -is	analyze	-is	Derivational (Noun Formation)	analyze
analytical	analysis + -ical	analyze	-ical	Derivational (Adjective Formation)	analyze

Explanation

- **Analyzing** represents an ongoing action and uses the **-ing** inflectional suffix.
- **Analysis** changes the verb into a noun.
- **Analytical** changes the noun into an adjective.

Final Normalized Representation

analyzing → *analyze*

analysis → *analyze*

analytical → *analyze*

Conclusion

All three words are indexed under the common root **analyze**, improving retrieval accuracy and allowing the search engine to match related queries.

Q2. Morphological Parsing of Disagree, Agreement, Agreeable

Scenario Understanding

A sentiment analysis system encounters different forms of the word **agree**. Morphological parsing helps identify prefixes, suffixes, and semantic changes to improve sentiment interpretation.

Morphological Parsing Strategy

- 1. Detect prefixes.
- 2. Identify suffixes.
- 3. Extract the base/root word.
- 4. Determine derivational changes.
- 5. Normalize to the common base.

Morphological Analysis

Word	Morphological Structure	Root	Prefix	Suffix	Transformation	Normalized Form
disagree	dis + agree	agree	dis-	—	Prefix creates opposite meaning	agree
agreement	agree + -ment	agree	—	-ment	Verb → Noun	agree
agreeable	agree + -able	agree	—	-able	Verb → Adjective	agree

Semantic Effect

- **dis-** changes the meaning to the opposite.
- **-ment** converts the verb into a noun.
- **-able** converts the verb into an adjective.

Normalized Output

disagree → *agree*

agreement → *agree*

agreeable → *agree*

Conclusion

Normalization enables the NLP system to recognize that all three words originate from **agree**, while semantic changes are preserved during sentiment analysis.

Q3. Morphology-Based Normalization of Govern, Government, Governance

Scenario Understanding

A news aggregation system must group related terms to improve topic modeling and document clustering.

Morphological Normalization Process

- 1. Identify the root.
- 2. Remove derivational suffixes.
- 3. Convert all forms to a common base.
- 4. Store the normalized representation.

Morphological Analysis

Word	Morphological Structure	Root	Affix	Word Class	Normalized Form
govern	govern	govern	—	Verb	govern
government	govern + -ment	govern	-ment	Noun	govern
governance	govern + -ance	govern	-ance	Noun	govern

Derivational Layers

- Govern → *Base verb.*
- Government → *Derived noun using -ment.*
- Governance → *Derived noun using -ance.*

Final Normalization

govern → *govern*

government → *govern*

governance → *govern*

Conclusion

All three words are mapped to **govern**, enabling efficient clustering and topic modeling because documents using different word forms are recognized as discussing the same concept.

Q4. Morphological Parsing of Activate, Activation, Reactivation

Scenario Understanding

A document classification system processes related words to improve semantic grouping and indexing.

Morphological Parsing Strategy

1. Identify prefixes.
2. Identify suffixes.
3. Extract the root.
4. Determine derivational layers.
5. Normalize to a common base.

Morphological Analysis

Word	Morphological Structure	Root	Prefix	Suffix	Transformation	Normalized Form
activate	activate	activate	—	—	Base verb	activate
activation	activate + -ion	activate	—	-ion	<i>Verb</i> → <i>Noun</i>	activate
reactivation	re + activate + -ion	activate	re-	-ion	Prefix + Noun Formation	activate

Semantic Changes

- **Activation** refers to the process of activating.
- **Reactivation** indicates activation occurring again.
- **re-** adds the meaning "again."
- **-ion** changes the verb into a noun.

Unified Base Representation

activate → *activate*

activation → *activate*

reactivation → *activate*

Conclusion

All three forms are indexed under **activate**, while preserving semantic information introduced by the prefix **re-** and suffix **-ion**.

Q5. Morphological Normalization of Create, Creates, Creating

Scenario Understanding

A search optimization system must treat grammatical variations as equivalent to improve retrieval performance.

Morphological Normalization Approach

1. Detect inflectional suffixes.
2. Remove tense and aspect markers.
3. Recover the base form.
4. Store the lemma for indexing.

Morphological Analysis

Word	Morphological Structure	Root	Suffix	Transformation	Normalized Form
create	create	create	—	Base Verb	create
creates	create + -s	create	-s	Third Person Singular	create
creating	create + -ing	create	-ing	Present Participle	create

Inflectional Transformations

- **-s** indicates third-person singular present tense.
- **-ing** indicates continuous/progressive aspect.
- These suffixes do **not** change the meaning of the base word.

Normalized Output

create → *create*

creates → *create*

creating → *create*

Conclusion

All grammatical variations are normalized to **create**, ensuring consistent indexing and retrieval while preserving the original meaning.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____